BIOCHEMISTRY CUMULATIVE EXAM

September 17, 1994

Protein structure has traditionally been subdivided into the categories of primary, secondary, tertiary and quaternary structure.

1. (20 points) Define the terms, "primary," "secondary," "tertiary," and "quaternary;" and give some reasons for using these subdivisions.

2. (30 points) Outline the necessary steps to determine the complete primary structure of a protein. Indicate one appropriate chemical reaction that is used in the process.

3. (30 points) Discuss two methods for obtaining information about the secondary structure of a protein. One of these should be a method that yields information about where particular elements of secondary structure are positioned within the primary structure, while the other may be a more general method. Compare the information that is derived from the two methods, and evaluate the relative effectiveness of each method.

4. (20 points) Explain two examples where protein quaternary structure is important for biological function.